

CONTINUIDAD

62) Determinar si la función $y = \frac{2x+1}{2x^2+x}$ es continua en $x=0$

$$\lim_{x \rightarrow 0} \frac{2x+1}{2x^2+x} = \lim_{x \rightarrow 0} \frac{2x+1}{x(2x+1)} = \lim_{x \rightarrow 0} \frac{1}{x} = \frac{1}{0} = \boxed{\infty}$$

ES DISCONTINUA
NO EVITABLE

63) $f(x) = \left(x \cdot \frac{\sin 5}{x}\right)$ en $x=0$

$$\begin{aligned} \lim_{x \rightarrow 0} \left(x \cdot \frac{\sin 5}{x}\right) &= \lim_{x \rightarrow 0} \frac{\sin 5}{\frac{1}{x}} = \lim_{x \rightarrow 0} \frac{\sin 5}{\frac{5 \cdot \frac{1}{5}}{x}} = \\ &= \frac{1}{\frac{1}{5}} \left(\lim_{x \rightarrow 0} \frac{\sin 5}{\frac{5}{x}} \right) = 5 \cdot 1 = \boxed{5} \end{aligned}$$

↓
LÍMITE NOTABLE

DISCONTINUA
EVITABLE

64) $f(x) = \left(\frac{x^2+3x-10}{x^2-11x+18}\right)$ en $x=2$

Numerador: $x = \frac{-3 \pm \sqrt{9+40}}{2} = \frac{-3 \pm 7}{2} \rightarrow \begin{cases} x_1 = 2 \\ x_2 = -5 \end{cases}$

Denominador: $x = \frac{11 \pm \sqrt{121-72}}{2} = \frac{11 \pm 7}{2} \rightarrow \begin{cases} x_1 = 9 \\ x_2 = 2 \end{cases}$

$$\lim_{x \rightarrow 2} \frac{(x-2)(x+5)}{(x-9)(x-2)} = \lim_{x \rightarrow 2} \frac{x+5}{x-9} = \frac{2+5}{2-9} = \frac{7}{-7} = \boxed{-1}$$

DISCONTINUA
EVITABLE

$$65) f(x) = \frac{x^2 + 2}{x - 1} \text{ en } x = 3$$

$$\lim_{x \rightarrow 3} \frac{x^2 + 2}{x - 1} = \frac{3^2 + 2}{3 - 1} = \frac{11}{2} \text{ CONTINUA}$$

$$66) f(x) = \frac{x^2}{1 - \cos^2 x} \text{ en } x = 0$$

$$\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^2 x} = \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x} = \frac{0}{0} \text{ indeterminado}$$

por trigonometría
 $\sin^2 x + \cos^2 x = 1$
 $\sin^2 = 1 - \cos^2 x$

$$\begin{aligned} & \downarrow \\ & \sin^2 x \\ & = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin^2 x}{x^2}} = \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}} = \frac{1}{\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2} = \frac{1}{\left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2} = \frac{1}{1^2} = \boxed{1} \end{aligned}$$

DISCONTINUA
EVITABLE

Resolver los siguientes ejercicios:

$$67) f(x) = \frac{3x + 2}{3x - 1} \text{ en } x = \infty$$

$$\lim_{x \rightarrow \infty} \frac{3x + 2}{3x - 1} = 1 \quad \text{DISCONTINUA EVITABLE}$$

$$68) f(x) = \left(\frac{\sin x}{\cos^2 x} \right) \text{ en } x = \frac{\pi}{6} \rightarrow (\text{Phi}) \quad \pi = 3,14 \text{ en radianes}$$

$$\lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin x}{\cos^2 x} = 0,66 \quad \text{CONTINUA}$$

$$69) f(x) = \frac{\sin^2 x}{\sqrt{\frac{x}{\pi}}} \text{ en } x = \frac{\pi}{4}$$

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin^2 x}{\sqrt{\frac{x}{\pi}}} = \boxed{1} \quad \text{CONTINUA}$$

RESOLUCIÓN EJERCICIOS DE CONTINUIDAD.

$$\begin{aligned} 67) \lim_{x \rightarrow \infty} \frac{3x+2}{3x-1} &= \lim_{x \rightarrow \infty} \frac{x(3+\frac{2}{x})}{x(3-\frac{1}{x})} = \lim_{x \rightarrow \infty} \frac{(3+\frac{2}{x})}{(3-\frac{1}{x})} = \frac{\lim_{x \rightarrow \infty} (3+\frac{2}{x})}{\lim_{x \rightarrow \infty} (3-\frac{1}{x})} \\ &= \frac{3+\frac{2}{\infty}}{3-\frac{1}{\infty}} = \frac{3+0}{3-0} = \frac{3}{3} = 1 \quad \text{DISCONTINUA EVITABLE.} \end{aligned}$$

$$\begin{aligned} 70) \lim_{x \rightarrow 2} \frac{x-2}{\sqrt{x^2-4}} &= \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{x^2-4})}{(\sqrt{x^2-4})(\sqrt{x^2-4})} = \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{x^2-4})}{(\sqrt{x^2-4})^2} = \\ &= \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{x^2-4})}{x^2-4} \quad x^2-4 = (x-2)(x+2) \\ &= \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{x^2-4})}{(x-2)(x+2)} = \frac{\sqrt{2^2-4}}{2+2} = \frac{0}{4} = 0 \quad \text{DISCONTINUA EVITABLE.} \end{aligned}$$

$$71) \lim_{x \rightarrow \infty} \frac{3^x - 3^{-x}}{3^x + 3^{-x}} = \text{dividimos todo por "3^x":}$$

$$\begin{aligned} &= \lim_{x \rightarrow \infty} \frac{\frac{3^x}{3^x} - \frac{3^{-x}}{3^x}}{\frac{3^x}{3^x} + \frac{3^{-x}}{3^x}} = \lim_{x \rightarrow \infty} \frac{1 - \frac{1}{3^x \cdot 3^x}}{1 + \frac{1}{3^x \cdot 3^x}} = \lim_{x \rightarrow \infty} \frac{1 - \frac{1}{3^{2x}}}{1 + \frac{1}{3^{2x}}} = \frac{\lim_{x \rightarrow \infty} (1 - \frac{1}{3^{2x}})}{\lim_{x \rightarrow \infty} (1 + \frac{1}{3^{2x}})} \\ &= \frac{1 - \frac{1}{3^\infty}}{1 + \frac{1}{3^\infty}} = \frac{1 - \frac{1}{\infty}}{1 + \frac{1}{\infty}} = \frac{1-0}{1+0} = \frac{1}{1} = 1 \quad \text{DISCONTINUA EVITABLE.} \end{aligned}$$

$$\begin{aligned} 72) \lim_{x \rightarrow -1} \frac{x^3 + x^2 - x - 1}{x^2 - x - 2} &= \text{FACTOREAMOS NUMERADOR Y DENOMINADOR} = \\ &= \lim_{x \rightarrow -1} \frac{(x^2-1)(x+1)}{(x-2)(x+1)} = \frac{(-1)^2-1}{-1-2} = \frac{1-1}{-3} = \frac{0}{-3} = 0 \quad \text{DISCONTINUA EVITABLE.} \end{aligned}$$